

National Curriculum of Pakistan 2022-2023
Assessment Framework Physics Grade-XI (HSSC-I)
Details of Content Areas/ SLOs

NCP SLOs Description	Form of Assessment	Cognitive Level	Remarks	Number of Periods Required (1 period=40 minutes)
Domain: A Measurements				
[SLO: P-11-A-01] Make reasonable estimates of physical quantities [of those quantities that are discussed in the topics of this grade]	Formative	Understanding	Question will not be asked in final examination	07 periods
[SLO: P-11-A-02] Express derived units as products or quotients of the SI base units	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-A-03] Analyze the homogeneity of physical equations [Through dimensional analysis]	Summative	Application	Question(s) will be asked in final examination.	
[SLO: P-11-A-04] Derive formulae in simple cases [Through using dimensional analysis]	Summative	Application	Question(s) will be asked in final examination.	
[SLO: P-11-A-05] Analyze and critique the accuracy and precision of data collected by measuring instruments	Summative	Application	Question(s) will be asked in final examination.	05 periods
[SLO: P-11-A-06] Assess the uncertainty in a derived quantity [By simple addition of absolute, fractional or percentage uncertainties]	Summative	Application	Question(s) will be asked in final examination.	
[SLO: P-11-A-07] Justify why all measurements contain some uncertainty	Summative	Understanding + Application	Question(s) will be asked in final examination.	
Domain: B Mechanics			Question(s) will be asked in final examination.	10 periods
[SLO: P-11-B-01] Represent a vector in 2-D as two perpendicular components	Summative	Understanding + Application	Question(s) will be asked in final examination.	
[SLO: P-11-B-02] Describe the product of two vectors (dot and cross-product) along with their properties.	Summative	Understanding + Application	Question(s) will be asked in final examination.	
[SLO: P-11-B-03] Derive the equations of motion [For uniform acceleration cases only. Derive from the definitions of velocity and acceleration as well as graphically].	Formative	Understanding	Question will not be asked in final examination	
[SLO: P-11-B-04] Solve problems using the equations of motion [For the cases of	Summative	Application	Question(s) will be asked in	

NCP SLOs Description	Form of Assessment	Cognitive Level	Remarks	Number of Periods Required (1 period=40 minutes)
uniformly accelerated motion in a straight line, including the motion of bodies falling in a uniform gravitational field without air resistance. This also includes situations where the equations of motion need to be resolved into vertical and horizontal components for 2-D motion]			final examination.	
[SLO: P-11-B-05] Evaluate and analyze projectile motion in the absence of air resistance [This includes solving problems making use of the below facts: (i) Horizontal component (V_H) of velocity is constant. (i) Acceleration is in the vertical direction and is the same as that of a vertically free falling object. (ii) The horizontal motion and vertical motion are independent of each other. Situations may require students to determine for projectiles: - How high does it go? - How far would it go along the level land? - Where would it be after a given time? - How long will it remain in flight? Situations may also require students to calculate for a projectile launched from ground height the - launch angle that results in the maximum range. - relation between the launch angles that result in the same range.] [SLO: P-11-B-06] Predict qualitatively how air resistance affects projectile motion [This includes analysis of both the horizontal component and vertical component of velocity and hence predicting qualitatively the range of the projectile.]	Summative	Application	Question(s) will be asked in final examination.	
[SLO: P-11-B-06] Predict qualitatively how air resistance affects projectile motion [This includes analysis of both the horizontal component and vertical component of velocity and hence predicting qualitatively the range of the projectile.]	Summative	Understanding + Application	Question(s) will be asked in final examination.	
[SLO: P-11-B-07] Express angles in radians	Summative	Knowledge+ Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-B-08] Define and calculate angular displacement, angular velocity and angular acceleration [This involves use of $S = r\theta$, $v = r\omega$, $\omega = 2\pi/T$, $a = r\omega^2$ and $a = v^2/r$ to solve problems].	Summative	Knowledge+ Understanding + Application	Question(s) will be asked in final examination.	05 periods
[SLO: P-11-B-09] Use equations of angular motion to solve problems involving rotational motions.	Summative	Application	Question(s) will be asked in final examination.	
[SLO: P-11-B-10] Analyze qualitatively motion in a curved path due to a perpendicular force.	Summative	Understanding + Application	Question(s) will be asked in final examination.	

NCP SLOs Description	Form of Assessment	Cognitive Level	Remarks	Number of Periods Required (1 period=40 minutes)
[SLO: P-11-B-11] Apply the principle of conservation of momentum to solve simple problems [Including elastic and inelastic interactions between objects in both one and two dimensions. Knowledge of the concept of coefficient of restitution is not required. Examples of applications include: - karate chops to break a pile of bricks - car crashes - ball & bat - the motion under thrust of a rocket in a straight line considering short thrusts during which the mass remains constant]	Formative	Application	Question will not be asked in final examination.	05 periods
[SLO: P-11-B-12] Predict and analyze motion for elastic collisions [This include making use of the fact that for an elastic collision, total kinetic energy is conserved and the relative speed of approach is equal to the relative speed of separation]	Summative	Application	Question(s) will be asked in final examination.	
[SLO: P-11-B-13] Justify why though the momentum of a closed system is always conserved, some change in kinetic energy may take place	Summative	Application	Question(s) will be asked in final examination.	
[SLO: P-11-B-14] Define and calculate centripetal force [Use $F = mr\omega^2$, $F = mv^2/r$]	Summative	Knowledge+ Application	Question(s) will be asked in final examination.	15 periods
[SLO: P-11-B-15] Analyze situations involving circular motion in terms of centripetal force [e.g. situations in which centripetal acceleration is caused by a tension force, a frictional force, a gravitational force, or a normal force.]	Summative	Application	Question(s) will be asked in final examination.	
[SLO: P-11-B-16] Explain why the objects in orbiting satellites appear to be weightless.	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-B-17] Describe how artificial gravity is created to counter weightlessness.	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-B-18] Define and calculate moment of inertia of a body and angular momentum.	Summative	Knowledge+ Application	Question(s) will be asked in final examination.	
[SLO: P-11-B-19] Derive and apply the relation between torque, moment of inertia and angular acceleration.	Summative	Application	Question(s) will be asked in final examination.	
[SLO: P-11-B-20] State and apply the law of conservation of angular momentum. Illustrate the applications of conservation of angular momentum in real life [Such as by flywheels to store rotational energy, by gyroscopes in navigation systems, by ice skaters to adjust their angular velocity].	Summative	Knowledge+ Understanding + Application	Question(s) will be asked in final examination.	

NCP SLOs Description	Form of Assessment	Cognitive Level	Remarks	Number of Periods Required (1 period=40 minutes)
[SLO: P-11-B-21] Justify how a centrifuge is used to separate materials using centripetal force	Summative	Application	Question(s) will be asked in final examination.	32 periods
[SLO: P-11-B-22] Distinguish between the structures of crystalline, glassy, amorphous, and polymeric solids.	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-B-23] Describe that deformation of solids in one dimension [that it is caused by a force and that in one dimension, the deformation can be tensile or compressive].	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-B-24] Define and use the terms stress, strain and the Young modulus	Summative	Application	Question(s) will be asked in final examination.	
[SLO: P-11-B-25] Describe an experiment to determine the Young modulus of a metal wire.	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-B-26] Describe and use the terms elastic deformation, plastic deformation and elastic limit	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-B-27] Justify why and apply the fact that the area under the force–extension graph represents the work done	Summative	Application	Question(s) will be asked in final examination.	
[SLO: P-11-B-28] Determine the elastic potential energy of a material [that is deformed within its limit of proportionality from the area under the force-extension graph. Also state and use $E_p = \frac{1}{2}Fx = \frac{1}{2}kx^2$ for a material deformed within its limit of proportionality].	Summative	Application	Question(s) will be asked in final examination.	
[SLO: P-11-B-29] Derive the formula for kinetic Energy [using the equations of motion]	Summative	Application	Question(s) will be asked in final examination.	
[SLO: P-11-B-30] Deduce the work done from force-displacement graph	Summative	Application	Question(s) will be asked in final examination.	
[SLO: P-11-B-31] Differentiate between conservative and non-conservative forces	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-B-32] Utilize the work – energy theorem in a resistive medium to solve problems.	Summative	Application	Question(s) will be asked in final examination.	
[SLO: P-11-B-33] Justify and use Archimedes' principle of flotation	Summative	Application	Question(s) will be asked in final examination.	
[SLO: P-11-B-34] Justify how ships are engineered to float in the sea	Summative	Application	Question(s) will be asked in final examination.	

NCP SLOs Description	Form of Assessment	Cognitive Level	Remarks	Number of Periods Required (1 period=40 minutes)
[SLO: P-11-B-35] Define and apply the terms: steady (streamline or laminar) flow, incompressible flow and non-viscous flow as applied to the motion of an ideal fluid	Summative	Knowledge+ Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-B-36] Use equation of continuity to solve problems	Summative	Application	Question(s) will be asked in final examination.	
[SLO: P-11-B-37] Explain that squeezing the end of a rubberpipe results in increase in flow velocity	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-B-38] Justify that the continuity is a form of the principle of conservation of mass.	Summative	Application	Question(s) will be asked in final examination.	
[SLO: P-11-B-39] Justify that the pressure difference can arise from different rates of flow of a fluid [Bernoulli effect].	Formative	Application	Question(s) will not be asked in final examination.	
[SLO: P-11-B-40] Explain and apply Bernoulli's equation for horizontal and vertical fluid flow	Summative	Application	Question(s) will be asked in final examination.	
[SLO: P-11-B-41] Explain why real fluids are viscous fluids.	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-B-42] Describe how viscous forces in a fluid cause a retarding force on an object moving through it.	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-B-43] Describe super fluidity [As the state in which a liquid will experience zero viscosity. Students should know the implications of this state e.g. this allows for super fluids to creep over the walls of containers to 'empty' themselves. It also implies that if you stir a superfluid, the vortices will keep spinning indefinitely.]	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-B-44] Analyze the real world applications of the Bernoulli effect [For example, atomisers in perfume bottles, the swinging trajectory of a spinning cricket ball and the lift of a spinning golf ball (the magnus effect), the use of Venturi ducts in filter pumps and car engines to adjust the flow of fluid, etc]	Summative	Application	Question(s) will be asked in final examination.	
Domain C: Heat and Thermodynamics				30 periods
[SLO: P-12-C-01] State that regions of equal temperature are in thermal equilibrium	Summative	Knowledge	Question(s) will be asked in final examination.	
[SLO: P-11-C-02] Relate a rise in temperature of an object to an increase in its internal energy	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-C-03] Apply the equation of state for an ideal gas [expressed as $pV = nRT$, where n = amount of substance (number of moles) and as $pV = NkT$, where	Summative	Application	Question(s) will be asked in final examination.	

NCP SLOs Description	Form of Assessment	Cognitive Level	Remarks	Number of Periods Required (1 period=40 minutes)
N = number of molecules]				
[SLO: P-11-C-04] State that the Boltzmann constant k is given by $k=R/N_A$	Summative	Knowledge	Question(s) will be asked in final examination.	
[SLO: P-11-C-05] Describe the basic assumptions of the kinetic theory of gasses [Including understanding the temperature, pressure and density conditions under which an ideal gas is a good approximation of a real gas.]	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-C-06] Use $W = p\Delta V$ for the work done when the volume of a gas changes at constant pressure.	Summative	Application	Question(s) will be asked in final examination.	
[SLO: P-11-C-07] Describe the difference between the work done by a gas and the work done on a gas.	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-C-08] Define and use the first law of thermodynamics [$Q = \Delta U + W$ expressed in terms of the increase in internal energy, the heating of the system (energy transferred to the system by heating) and the work done on the system]	Summative	Knowledge + Understanding + Application	Question(s) will be asked in final examination.	
[SLO: P-11-C-09] Explain qualitatively, in terms of particles, the relationship between the pressure, temperature and volume of a gas [Specifically the below case: (a) pressure and temperature at constant volume. (b) volume and temperature at constant pressure (c) pressure and volume at a constant temperature]	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-C-10] Use the equation, including a graphical representation of the relationship between pressure and volume for a gas at constant temperature.	Summative	Application	Question(s) will be asked in final examination.	
[SLO: P-11-C-11] Justify how the first law of thermodynamics expresses the conservation of energy.	Summative	Application	Question(s) will be asked in final examination.	
[SLO: P-11-C-12] Relate a rise in temperature of a body to an increase in its internal energy.	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-C-13] State the working principle of a heat engine.	Summative	Knowledge	Question(s) will be asked in final examination.	
[SLO: P-11-C-14] Describe the concept of reversible and irreversible processes.	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-C-15] State and explain the second law of thermodynamics.	Summative	Understanding	Question(s) will be asked in final examination.	

NCP SLOs Description	Form of Assessment	Cognitive Level	Remarks	Number of Periods Required (1 period=40 minutes)
[SLO: P-11-C-16] State the working principle of Carnot’s engine	Summative	Knowledge+ Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-C-17] Describe that refrigerator is a heat engine operating in reverse as that of an ideal heat engine.	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-C-18] Explain that an increase in temperature increases the disorder of the system.	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-C-19] Explain that increase in entropy means degradation of energy.	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-C-20] Explain that energy is degraded during all natural processes.	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-C-21] Identifying that system tends to become less orderly over time.	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-C-22] Explain that Entropy, S, is a thermodynamic quantity that relates to the degree of disorder of the particles in a system.	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-C-23] State that the Carnot cycle sets a limit for the efficiency of a heat engine at the temperatures of its heat reservoirs given by Efficiency=1- T(cold reservoir)/T(hot reservoir)	Summative	Knowledge	Question(s) will be asked in final examination.	
Domain D: Waves				
[SLO: P-11-D-01] Use intensity = power/area to solve problems. Use Intensity $\propto$ (amplitude) ² for a progressive wave to solve problems	Summative	Application	Question(s) will be asked in final examination.	26 periods
[SLO: P-11-D-02] Explain that when a source of sound waves moves relative to a stationary observer, the observed frequency is different from the source frequency [describing of the Doppler effect for a stationary source and a moving observer is not required]	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-D-03] Use the expression $f_o = (f_s \cdot v)/(v \pm v_s)$ for the observed frequency when a source of sound waves moves relative to a stationary observer.	Summative	Application	Question(s) will be asked in final examination.	
[SLO: P-11-D-04] Explain the applications of the Doppler effect [such as radar, sonar, astronomy, satellite, radar speed traps and studying cardiac problems in	Summative	Understanding	Question(s) will be asked in final examination.	

NCP SLOs Description	Form of Assessment	Cognitive Level	Remarks	Number of Periods Required (1 period=40 minutes)
humans]				
[SLO: P-11-D-05] Explain that polarization is a phenomenon associated with transverse waves	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-D-06] Define and apply Malus's law [$I = I_0 \cos^2 \theta$ to calculate the intensity of a plane-polarized electromagnetic wave after transmission through a polarizing filter or a series of polarizing filters. (calculation of the effect of a polarizing filter on the intensity of an unpolarized wave is not required).]	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-D-07] Use the principle of superposition of waves to solve problems	Summative	Application	Question(s) will be asked in final examination.	
[SLO: P-11-D-08] Differentiate between constructive and destructive interference.	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-D-09] Apply the principle of superposition to explain the working of noise canceling headphones.	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-D-10] Illustrate experiments that demonstrate stationary waves [using microwaves, stretched strings and air columns (it will be assumed that end corrections are negligible; knowledge of the concept of end corrections is not required)]	Formative	Application	Question(s) will not be asked in final examination.	
[SLO: P-11-D-11] Explain the formation of a stationary wave using graphical representation	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-D-12] Explain the formation of harmonics in stationary waves.	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-D-13] Analyze experiments that demonstrate diffraction [including the qualitative effect of the gap width relative to the wavelength of the wave; for example diffraction of water waves in a ripple tank]	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-D-14] Explain the term coherence.	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-D-15] Explain beats [as the pulsation caused by two waves of slightly different frequencies interfering with each other]	Summative	Knowledge + Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-D-16] Illustrate examples of how beats are generated in musical	Summative	Understanding	Question(s) will be asked in	

NCP SLOs Description	Form of Assessment	Cognitive Level	Remarks	Number of Periods Required (1 period=40 minutes)
instruments		+ Application	final examination.	
[SLO: P-11-D-17] Explain the use of polaroids in sky photography and stress analysis of materials	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-D-18] Describe qualitatively gravitational waves [as waves of the intensity of gravity generated by the accelerated masses of an orbital binary system that propagate as waves outward from their source at the speed of light]	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-D-19] State that as a gravitational wave passes a body with mass the distortion in space-time can cause the body to stretch and compress periodically	Formative	Knowledge	Question(s) will not be asked in final examination.	
[SLO: P-11-D-20] State that gravitational waves pass through the Earth due to far off celestial events, but they are very minute amplitude	Formative	Knowledge	Question(s) will not be asked in final examination.	
[SLO: P-11-D-21] Describe the use of interferometers in detecting gravitational waves [Interferometers are very sensitive detection devices that make use of the interference of laser beams (working and set up details are not required) and were used to first detect the existence of gravitational waves]	Formative	Understanding	Question(s) will not be asked in final examination.	
Domain E: Electricity and Magnetism				32 periods
[SLO: P-11-E-01] State that an electric field is an example of a field of force	Summative	Knowledge	Question(s) will be asked in final examination.	
[SLO: P-11-E-02] Define and calculate electric field strength [Use $F = qE$ for the force on a charge in an electric field. Use $E = \frac{\Delta V}{\Delta d}$ to calculate the field strength of the uniform field between charged parallel plates]	Summative	Knowledge+ Understanding + Application	Question(s) will be asked in final examination.	
[SLO: P-11-E-03] Represent an electric field by means of field lines	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-E-04] Describe the effect of a uniform electric field on the motion of charged particles	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-E-05] State that, for a point outside a spherical conductor, the charge on the sphere may be considered to be a point charge at its center	Summative	Knowledge	Question(s) will be asked in final examination.	
[SLO: P-11-E-06] Explain how a Faraday cage works [by inducing internal electric fields that work to shield the inside from the influence of external electric fields]	Summative	Understanding	Question(s) will be asked in final examination.	

NCP SLOs Description	Form of Assessment	Cognitive Level	Remarks	Number of Periods Required (1 period=40 minutes)
SLO: P-11-E-07] State and apply Coulomb's law [$F = k \frac{Q_1 Q_2}{r^2}$ for the force between two point charges in free space, where $k = \frac{1}{4\pi\epsilon_0}$]	Summative	Application	Question(s) will be asked in final examination.	
[SLO: P-11-E-08] Use $E = k \frac{Q}{r^2}$ for the electric field strength due to a point charge in free space.	Summative	Application	Question(s) will be asked in final examination.	
[SLO: P-09-E-09] Describe how Ferro fluids work [they make use of temporary soft magnetic materials suspended in liquids to develop fluids that react to the poles of a magnet and have many applications in fields such as electronics]	Formative	Understanding	Question(s) will not be asked in final examination.	
[SLO: P-09-E-10] Use, for a current-carrying conductor, the expression $I = Anvq$ [where n is the number of charge carriers per unit volume.]	Summative	Understanding + Application	Question(s) will be asked in final examination.	
[SLO: P-09-E-11] State and use $V = W/Q$	Summative	Understanding + Application	Question(s) will be asked in final examination.	
[SLO: P-09-E-12] State and use $P = IV$, $P = I^2 R$ and $P = V^2/R$	Summative	Understanding + Application	Question(s) will be asked in final examination.	
[SLO: P-09-E-13] State and use $R = \rho L/A$	Summative	Understanding + Application	Question(s) will be asked in final examination.	
[SLO: P-09-E-14] State that the resistance of a light-dependent resistor (LDR) decreases as the light intensity increases	Summative	Knowledge+ Understanding	Question(s) will be asked in final examination.	
[SLO: P-09-E-15] Define and use the electromotive force (e.m.f.) [of a source as energy transferred per unit charge in driving charge around a complete circuit]	Summative	Application	Question(s) will be asked in final examination.	
[SLO: P-09-E-16] Distinguish between e.m.f. and potential difference (p.d.) in terms of energy considerations	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-09-E-17] Explain the effects of the internal resistance of a source of e.m.f. on the terminal potential difference	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-09-E-18] State Kirchhoff's first law and describe that it is a consequence	Summative	Knowledge	Question(s) will be asked in	

NCP SLOs Description	Form of Assessment	Cognitive Level	Remarks	Number of Periods Required (1 period=40 minutes)
of conservation of charge			final examination.	
[SLO: P-09-E-19] State Kirchhoff's second law and describe that it is a consequence of conservation of Energy	Summative	Knowledge + Understanding	Question(s) will be asked in final examination.	
[SLO: P-09-E-20] Derive, using Kirchhoff's laws, a formula for the combined resistance of two or more resistors in series	Summative	Application	Question(s) will be asked in final examination.	
[SLO: P-09-E-21] Derive and apply a formula for the combined resistance of two or more resistors in parallel	Summative	Application	Question(s) will be asked in final examination.	
[SLO: P-09-E-22] Use Kirchhoff's laws to solve simple circuit problems	Summative	Application	Question(s) will be asked in final examination.	
[SLO: P-09-E-23] State and use the principle of the potentiometer as a means of comparing potential differences	Summative	Understanding + Application	Question(s) will be asked in final examination.	
[SLO: P-09-E-24] Explain the use of a galvanometer in null methods	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-09-E-25] Explain the use of thermistors and light- dependent resistors in potential dividers [to provide a potential difference that is dependent on temperature and light intensity]	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-09-E-26] Explain the internal resistance of sources and its consequences for external circuits	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-09-E-27] Explain how inspectors can easily check the reliability of a concrete bridge with carbon fibers as the fibers conduct electricity	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-09-E-28] Define and explain magnetic fields	Summative	Knowledge	Question(s) will be asked in final examination.	18 periods
[SLO: P-09-E-29] State that a force might act on a current- carrying conductor placed in a magnetic field	Summative	Knowledge + Understanding	Question(s) will be asked in final examination.	
[SLO: P-09-E-30] Use the equation $F = BIL \sin(\theta)$ [with directions as interpreted by Fleming's left-hand rule to solve problems]	Summative	Application	Question(s) will be asked in final examination.	
[SLO: P-09-E-31] Define magnetic flux density [as the force acting per unit current per unit length on a wire placed at right angles to the magnetic field]	Summative	Knowledge	Question(s) will be asked in final examination.	

NCP SLOs Description	Form of Assessment	Cognitive Level	Remarks	Number of Periods Required (1 period=40 minutes)
[SLO: P-09-E-32] Use $F=Bqv \sin(\theta)$ to solve problems	Summative	Application	Question(s) will be asked in final examination.	
[SLO: P-09-E-33] Describe the motion of a charged particle moving in a uniform magnetic field perpendicular to the direction of motion of the particle	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-09-E-34] Explain how electric and magnetic fields can be used in velocity selection	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-09-E-35] Sketch magnetic field patterns due to the currents in a long straight wire, a flat circular coil and a long solenoid	Summative	Application	Question(s) will be asked in final examination.	
[SLO: P-09-E-36] State that the magnetic field due to the current in a solenoid is increased by a ferrous core.	Summative	Knowledge	Question(s) will be asked in final examination.	
[SLO: P-09-E-37] Explain the origin of the forces between current-carrying conductors and determine the direction of the forces.	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-09-E-38] Define magnetic flux [as the product of the magnetic flux density and the cross-sectional area perpendicular to the direction of the magnetic flux density]	Summative	Knowledge	Question(s) will be asked in final examination.	
[SLO: P-09-E-39] Use $\Phi=BA$ to solve problems	Summative	Application	Question(s) will be asked in final examination.	
[SLO: P-09-E-40] Use the concept of magnetic flux linkage	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-09-E-41] Explain experiments that demonstrate Faraday's and Lenz's laws [(a) that a changing magnetic flux can induce an e.m.f. in a circuit, (b) that the induced e.m.f. is in such a direction as to oppose the change producing it, (c) the factors affecting the magnitude of the induced e.m.f.]	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-09-E-42] Use Faraday's and Lenz's laws of electromagnetic induction to solve problems	Summative	Application	Question(s) will be asked in final examination.	
[SLO: P-09-E-43] Explain how seismometers make use of electromagnetic induction to the earthquake detection [specifically in terms of: (i) any movement or vibration of the rock on which the seismometer rests (buried in a protective case) results in relative motion between the magnet and the coil (suspended by a spring from the frame.) (ii) the emf induced in the coil is directly proportional	Summative	Understanding + Application	Question(s) will be asked in final examination.	

NCP SLOs Description	Form of Assessment	Cognitive Level	Remarks	Number of Periods Required (1 period=40 minutes)
to the displacement associated]				
Domain F: Modern Physics				08 periods
[SLO: P-11-F-01] Distinguish between inertial and non-inertial frames of reference.	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-F-02] Describe the significance of Einstein's assumption of the constancy of the speed of light.	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-F-03] Describe that if c is constant then space and time become relative.	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-F-04] State the postulates of Special relativity	Summative	Knowledge	Question(s) will be asked in final examination.	
[SLO: P-11-F-05] Explain qualitatively and quantitatively the consequences of special relativity [Specifically in the case of: a— the relativity of simultaneity. b— the equivalence between mass and energy. c— length contraction. d— time dilation. e— mass increase]	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-F-06] State that space-time is a mathematical model in relativity that treats time as a fourth dimension of the traditional three dimensions of space (It can be thought of as a metaphorical sheet of paper that can bend, and when it bends it can cause effects such as stretching and compression seen when gravitational waves pass through objects.)	Formative	Knowledge	Question(s) will not be asked in final examination.	
[SLO: P-11-F-07] State that nucleon number and charge are conserved in nuclear processes	Summative	Knowledge	Question(s) will be asked in final examination.	26 periods
[SLO: P-11-F-08] Describe the composition, mass and charge of α , β and γ radiations [both β^- (electrons) and β^+ (positrons) are included]	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-F-09] Explain that an antiparticle has the same mass but opposite charge to the corresponding particle [give the example that a positron is the antiparticle of an electron]	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-F-10] State that (electron) antineutrinos are produced during β^- decay and (electron) neutrinos are produced during β^+ decay	Summative	Knowledge	Question(s) will be asked in final examination.	

NCP SLOs Description	Form of Assessment	Cognitive Level	Remarks	Number of Periods Required (1 period=40 minutes)
[SLO: P-11-F-11] Explain that α -particles have discrete energies but that β -particles have a continuous range of energies because (anti)neutrinos are emitted in β -decay	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-F-12] Describe quarks and antiquarks (as a fundamental) [including that there are six flavors (types) of quark: up, down, strange, charm, top and bottom]	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-F-13] Describe protons and neutrons in terms of their quark composition	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-F-14] State that a hadron may be either a baryon (consisting of three quarks) or a meson (consisting of one quark and an antiquark)	Summative	Knowledge	Question(s) will be asked in final examination.	
[SLO: P-11-F-15] Describe the changes to quark composition that take place during β^- and β^+ decay	Summative	Knowledge	Question(s) will be asked in final examination.	
[SLO: P-11-F-16] State that electrons and neutrinos are fundamental particles called leptons	Summative	Knowledge+ Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-F-17] State, W, Z, gluon, and photons as fundamental particles called exchange particles or force carriers	Summative	Knowledge+ Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-F-18] State the Higgs Boson as a fundamental particle which is responsible for the particle's mass.	Summative	Knowledge+ Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-F-19] Explain that every subatomic particle has a corresponding antiparticle [that has the same mass as a given particle but opposite electric or magnetic properties according to the Standard Model of Particle Physics]	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-F-20] Describe protons and neutrons in terms of their quark composition	Summative	Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-F-21] State that a hadron may be either a baryon (consisting of three quarks) or a meson (consisting of one quark and an antiquark)	Summative	Knowledge+ Understanding	Question(s) will be asked in final examination.	
[SLO: P-11-F-22] Explain that there are various contending theories about what 'mass' and 'force' are generated from [e.g. that these are generated from quantum fields when they are energized, or from multidimensional 'strings' that	Formative	Understanding	Question(s) will not be asked in final examination.	

NCP SLOs Description	Form of Assessment	Cognitive Level	Remarks	Number of Periods Required (1 period=40 minutes)
vibrate in higher dimensions to give rise to particles (no further technical knowledge beyond these simple descriptions is expected at this level)]				
[SLO: P-11-F- B] Explain the working principle of particle accelerators and also their uses.	Formative	Understanding	Question(s) will not be asked in final examination.	
[SLO: P-11-F- 4] Explain that antimatter is the counterpart of matter [e.g. a positron is the antimatter counterpart to an electron]	Formative	Understanding	Question(s) will not be asked in final examination.	
[SLO: P-11-F- A] Illustrate that antiparticles usually have the same weight, but opposite charge, compared to their matter counterparts	Formative	Understanding	Question(s) will not be asked in final examination.	
[SLO: P-11-F- A] State that most of the matter in the observable universe is matter	Formative	Knowledge	Question(s) will not be asked in final examination.	
[SLO: P-11-F- L] Describe the asymmetry of matter and antimatter in the universe as an unsolved mystery	Formative	Understanding	Question(s) will not be asked in final examination.	
[SLO: P-11-F-28] Describe annihilation reactions [a particle meets its corresponding antiparticle, they undergo annihilation reactions in which either all the mass is converted to heat and light energy, or some mass is left over in the form of new subatomic particles.]	Summative	Understanding	Question(s) will be asked in final examination.	28 periods
Domain N: Experimentation Skills				
[SLO: P-11-N-01] - test that the lab equipment is functioning properly, without any potential risk of injury, before conducting an experiment	Formative for PBA	Application	Question will not be asked in final examination, however, it	
[SLO: P-11-N-02] - ensure that work space for conducting the experiment is not too crowded with apparatus as to be hazardous	Formative for PBA	Application	will be part of Lab work.	
[SLO: P-11-N-03] - ensure that safe distance is kept at all times from other investigators who may be handling lab apparatus	Formative for PBA	Application	Question will not be asked in final examination, however, it	
[SLO: P-11-N-04] - suggest broadly what potential bodily harm could occur from physical, chemical, biological and safety hazards in the context of the experiment being conducted	Formative for PBA	Application	will be part of Lab work.	
[SLO: P-11-N-05] - State that it is always better to ask for help from the lab instructor when unsure of how to use new apparatus	Formative for PBA	Application	Question will not be asked in final examination, however, it	
[SLO: P-11-N-06] Set up apparatus correctly without assistance from a supervisor	Formative for PBA	Application	Question will not be asked in final examination, however, it	